MyoGraph 3D: coronary artery disease probabilities on an interactive 3D heart

Track A, Multimodal AI Hackathon 2026: Cardiovascular Risk Visualization & Prediction. Model version `<!--v:784a6c5e-->xgb-v1-b60472ec<!--/v-->`*, trained 2026-10-13. Generated values come from the committed artifacts via* `scripts/report_numbers.py`*.*

1. Problem and approach

Track A asks for a system that predicts cardiovascular risk and makes it visible. MyoGraph 3D takes a patient's clinical record (demographics, history, examination, ECG, laboratory values, echocardiography) and returns model estimates of the probability of overall coronary artery disease (CAD) and of stenosis in each of 3 coronary arteries (LAD, LCX, RCA). Those probabilities are painted live onto a 3D heart, so the user sees which vessel the model considers most likely to be narrowed, and the number next to it.

Three ideas shape the design. First, honest validation: a fixed hold-out, repeated cross-validation on the training part only, and a split-stability analysis that shows how much one lucky or unlucky split can move a score. Second, calibrated, explainable output: every probability is calibrated and decomposes exactly into per-input contributions. Third, one source of truth: the feature schema and the target-to-mesh map live in two YAML files, and the backend, API, form and 3D scene all derive from them.

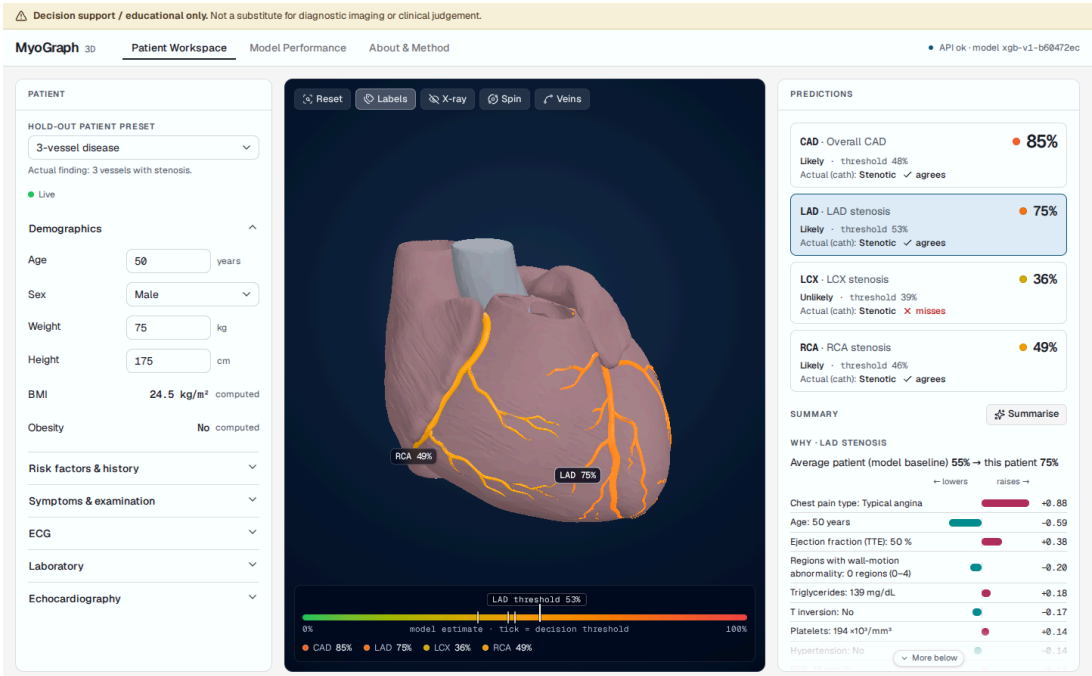


Figure 1. Patient Workspace with a three-vessel hold-out preset (`docs/img/workspace-3vessel.png`*).*

2. Data and preprocessing

Dataset. UCI *Extension of Z-Alizadeh Sani* (Alizadehsani, Roshanzamir & Sani, 2013; CC BY 4.0; doi:10.24432/C5461K): 303 patients, one row each, no missing values. Targets are overall CAD (catheterisation label) and stenosis of LAD, LCX, RCA. Positive rates in the training split: CAD 71%, LAD 59%, LCX 38%, RCA 38%.

Quirks handled in `config/features.yaml`.

- `Sex` contains the typo `Fmale`, mapped to female.
- Binary columns mix `Y`/`N` and `1`/`0`; each feature lists the raw values that mean yes.
- `VHD` has a lowercase `mild`; it becomes an ordinal 0–3.
- `Exertional CP` is constant and dropped.
- `BMI` and `Obesity` are derived from weight and height, server-side, and recomputed for training as well, so training and serving share one code path (this overrides one inconsistent `Obesity` row).
- Composites: smoking status and chest-pain type are mutually exclusive in the data, so each is one input that expands into several dataset columns; `BBB` is one-hot encoded.

Leakage exclusion. `Cath`, `LAD`, `LCX`, `RCA` are never model inputs for any target, including the other vessels when predicting one vessel. A test (`test_leakage.py`) enforces this from the schema.

Schema-driven. The schema holds 51 features: 22 number, 3 category, 2 derived, 22 binary, 2 composite. The client sends 49 inputs (derived fields are ignored if sent). No feature or vessel list is hard-coded in Python or TypeScript.

3. Model and validation

Model. One `XGBClassifier` per target behind a common `RiskModel` interface: 300 trees, depth 3, learning rate 0.05, subsample 0.8, column sample 0.8, `min_child_weight` 2, L2 1.0, and `scale_pos_weight` set per target to negatives/positives. The parameters are fixed baseline values; no hyper-parameter search was run.

Protocol.

1. Stratified 20% hold-out (seed 42), stratified on the number of stenotic vessels (0–3) so all four targets stay balanced: 242 train, 61 test.
2. Development metrics: 5-fold stratified CV repeated 3 times (seed 42) on the training set only. The hold-out is a final sanity check; it is never used for tuning or for CV.
3. Calibration: out-of-fold raw margins on train, then Platt scaling `logit = a·margin + b`; the final model is refit on all training rows. The served probability is `sigmoid(a·margin + b)`.

Target	a (slope)	b (intercept)
CAD	0.926	0.513
LAD	0.656	0.261
LCX	0.537	-0.162
RCA	0.358	-0.271

4. Decision thresholds: Youden's J maximiser on the training out-of-fold calibrated probabilities (never the hold-out). Inside each CV fold the threshold comes from that fold's own training predictions, so CV stays honest.

Results. CV is the primary figure (mean ± std over 15 folds); the hold-out has n = 61 and is noisy.

Target	Prevalence (train / test)	ROC-AUC CV	ROC-AUC hold-out	Brier CV	Brier hold-out
Overall CAD	71% / 72%	0.932 ± 0.030	0.824	0.085 ± 0.016	0.174
LAD stenosis	59% / 56%	0.835 ± 0.054	0.804	0.161 ± 0.022	0.183
LCX stenosis	38% / 44%	0.729 ± 0.080	0.783	0.203 ± 0.022	0.204
RCA stenosis	38% / 36%	0.698 ± 0.058	0.695	0.216 ± 0.011	0.205

Brier scores do not depend on the threshold. Threshold-dependent metrics at the tuned thresholds:

Target	Tuned threshold	CV spread of threshold	Recall CV	Recall hold-out	F1 CV	F1 hold-out	F1 hold-out at 0.5
Overall CAD	0.477	0.637 ± 0.068	0.904 ± 0.071	0.773	0.914 ± 0.034	0.810	0.810
LAD stenosis	0.535	0.510 ± 0.078	0.834 ± 0.140	0.824	0.795 ± 0.072	0.789	0.800
LCX stenosis	0.395	0.380 ± 0.049	0.690 ± 0.120	0.630	0.607 ± 0.064	0.654	0.522
RCA stenosis	0.462	0.349 ± 0.044	0.735 ± 0.128	0.500	0.588 ± 0.060	0.564	0.400

Youden thresholds are noisy at this sample size: the spread across CV folds is shown above, and the tuned value can lower or raise F1 relative to 0.5 (last column). Prevalence matters for reading these numbers: a constant predictor of the majority class would reach accuracy 72% on overall CAD in the hold-out, so Brier and ROC-AUC are the more informative figures.

Split stability. The hold-out ROC-AUC and Brier were recomputed over 20 split seeds (0–19; same stratification, same model seed). Seed 42 is the hardest split for overall CAD: its AUC is the lowest of the 20, and its Brier score is at percentile 100 (the worst).

Target	Seed-42 AUC	Median	IQR	Min – max	Seed-42 percentile
Overall CAD	0.824	0.911	0.874 – 0.924	0.836 – 0.969	0
LAD stenosis	0.804	0.810	0.794 – 0.861	0.756 – 0.931	35
LCX stenosis	0.783	0.772	0.720 – 0.791	0.642 – 0.807	60
RCA stenosis	0.695	0.681	0.656 – 0.729	0.588 – 0.773	65

The headline hold-out CAD AUC (0.824) is therefore a pessimistic draw; the median over seeds is 0.911. We kept seed 42 as the official split because it was fixed before the stability analysis, and report the spread instead of choosing a friendlier seed.

4. Interpretability

TreeSHAP in calibrated log-odds. At serve time the app uses XGBoost's native `pred_contribs`, which gives exact TreeSHAP contributions in margin space. They are scaled by the calibration slope `a`, and the bias becomes `a·bias + b`. Contributions plus base value therefore sum exactly to the calibrated logit, which `test_explain_additivity.py` verifies. Contributions of multi-column inputs (chest-pain type, smoking, bundle-branch block) are summed back to their input by additivity.

WHY panel. For the selected target, the top 10 inputs are shown as diverging bars (red raises the estimate, teal lowers it) in log-odds, with the header "average patient (model baseline) → this patient". Global importance on the training set:

Target	Top-3 inputs by mean |SHAP| (log-odds)
CAD	Chest pain type (1.49), Age (0.70), Ejection fraction (TTE) (0.59)
LAD	Chest pain type (0.68), Ejection fraction (TTE) (0.49), Age (0.44)
LCX	Age (0.42), Triglycerides (0.30), Chest pain type (0.29)
RCA	Chest pain type (0.21), Age (0.21), Diabetes mellitus (0.19)

Physiological breakdown. A table of numeric measurements, derived BMI/obesity and key binaries: value and unit, typical adult reference range (display only), status (low/normal/high), and signed contribution to the selected target. Values outside the training range raise a warning but are still scored.

Plain-language summary. On request, `POST /api/summary` re-runs the prediction server-side and builds a facts object (percentages, verdicts, top factors, range warnings). A deterministic template always works. If `GROQ_API_KEY` is set, only that facts object is sent to Groq to reword it (temperature 0, 8 s timeout, one retry on rate-limit). The output is validated: it must carry the disclaimer, stay near 120 words, contain no number absent from the facts, state every target's percentage, mention only factors from the selected target's list, and avoid advice or diagnostic phrasing. Any failure falls back to the template; the UI labels which source produced the text.

5. 3D pipeline

`scripts/build_heart_glb.py` turns BodyParts3D 4.0 (DBCLS) into `heart.glb`, driven by `config/anatomy.yaml`.

- **Mapping.** Version 4.0 ships `FJ<n>.obj` files, not FMA-named ones. The script indexes the zip by each file's header and joins to FMA concepts through the parts list. A plain FMA id is included if it resolves; a nested list means "first available alternative". Some ids are composites with no mesh of their own (their children carry the geometry); missing ids are logged, and a target vessel with no geometry fails the build loudly.
- **De-duplication.** Each source file may belong to only one node; the script fails if two nodes claim the same file.
- **Decimation.** Quadric decimation to a per-node triangle budget, then recentring on the heart centroid, scaling to about 2 scene units, and conversion to Y-up.
- **Extras.** Each node carries an `anchor` (a vertex on its surface for the label), a `viewDir` for rotate-to-vessel, and visibility flags.

Node	Triangles	Budget	Note
LAD	8000	8000	
LCX	3018	8000	
RCA	10000	10000	
LM	204	2000	
heart	40000	40000	
aorta	396	6000	optional
veins	6000	6000	hidden by default

Total 67,618 triangles (61,618 with the default-hidden veins off), against a build limit of 80,000; the file is 1.27 MB against a 3 MB limit. The build script enforces both.

Correspondence and colour. Each target in `features.yaml` maps to exactly one node in `anatomy.yaml` (tested); the frontend receives the node names from `/api/features` and never names a vessel. Colour is the calibrated probability on a green–amber–red scale interpolated in OKLCH; the number is always shown beside it, and the legend marks each target's tuned threshold. Selecting a card or vessel rotates the camera to that node's view direction and fits its bounding box. Overall CAD is bound to the heart body.

No-GPU design. `frameloop="demand"`, device pixel ratio at most 1.5, no shadows, no Draco decoder, selection by emissive boost, and the post-processing outline dropped when a frame-budget monitor reports slow frames.

6. System and architecture

Backend. FastAPI with Pydantic v2 models as the API contract. Endpoints: `/api/health`, `/api/features`, `/api/presets`, `/api/predict`, `/api/summary`, `/api/metrics`. The API loads committed artifacts at startup and never trains. Inputs are validated (bounds, non-finite values, missing fields, unknown options) with field-level 422 errors.

Frontend. React 19 and TypeScript (strict) with react-three-fiber for the scene. The form is generated from `/api/features`; predictions refresh on a 300 ms debounce with stale requests aborted. TypeScript API types are generated from the Pydantic schema (`make types`), never hand-written.

Extensibility. A new clinical feature is a `features.yaml` entry plus `make train`. A new model is a class implementing `RiskModel` plus a registry entry. A new anatomical structure or target is an `anatomy.yaml` entry plus `make assets`.

Operations and reproducibility. `make setup | dev | train | assets | types | check | docker`. `make docker` serves the app at http://localhost:8080 (nginx proxying to the API container). Seeds are fixed at 42; training is deterministic, and `model_card.json` records the data hash (`<!--v:995b6bef-->b60472ecebdd<!--/v-->…`) and git SHA. The backend test suite collects 63 tests across 9 files (anatomy, api, explain_additivity, health, leakage, preprocess, presets, split, summary); the main gates are leakage, SHAP additivity, split stratification and anatomy mapping.

7. Usage

1. Run `make docker` and open http://localhost:8080 (or `make setup && make dev` for the development servers on :5173 and :8000).
2. Load a preset from the dropdown. The presets are 6 patients taken only from the hold-out set (no-cad, lad-only, lcx-rca-only, 2-vessel, 3-vessel, hard-case), so the model never saw them in training; each shows the actual catheterisation result next to the model estimate.
3. Read the right panel top-down: overall CAD estimate, then the vessel cards, then the WHY bars for the selected target. Click a card or a vessel in the 3D view to change the selection. Edit any field and the estimates update live.
4. The Model Performance tab shows CV and hold-out metrics, ROC curves, confusion matrices, calibration and importance. The About tab holds the method and credits.

Colour is never the only signal: each percentage is printed, and "Likely/Unlikely" compares the estimate with that target's tuned threshold. The disclaimer banner is fixed on every tab.

8. Limitations and ethics

- **Small, single-centre data.** 303 patients from one hospital; the hold-out has 61. Confidence intervals are wide, and results may not transfer to other populations or devices.
- **Non-imaging inputs only.** The model never sees angiography or CT, so vessel-level estimates are population-pattern estimates, not anatomy. Localising stenosis is harder than detecting CAD: LCX and RCA hold-out ROC-AUC are 0.783 and 0.695, and their CV Brier scores (0.203, 0.216) improve only modestly on a prevalence-only predictor (0.236 and 0.236, from $p(1-p)$ on training prevalence).
- **Noisy thresholds and split sensitivity.** Tuned thresholds move by ±0.08 or more between folds; hold-out AUC for overall CAD ranges from 0.836 to 0.969 across split seeds.
- **Models can disagree.** Overall CAD and each vessel are separate models; the interface notes this when verdicts conflict.
- **Not clinically validated.** No prospective or external validation. The tool is for education and decision support, not a substitute for imaging or clinical judgement, and it gives no treatment advice. The wording throughout is "probability" or "model estimate".
- **Groq data note.** The optional wording feature is off unless a key is configured. When on, the facts object (percentages, verdicts, and factor labels with values) is sent to Groq's API; raw input records are not. No key means no external call.

9. Credits and licenses

- Dataset: Alizadehsani, R., Roshanzamir, M., & Sani, Z. (2013). *extention of Z-Alizadeh sani dataset*. UCI Machine Learning Repository. https://doi.org/10.24432/C5461K (CC BY 4.0).
- 3D anatomy: "BodyParts3D, © The Database Center for Life Science licensed under CC Attribution 4.0 International". The source OBJ headers still embed CC BY-SA 2.1 Japan, so the derived `heart.glb` is released under CC BY-SA 4.0.
- Code: MIT. Libraries: XGBoost, scikit-learn, pandas, FastAPI, Pydantic, React, three.js and react-three-fiber, Tailwind, shadcn/ui, Recharts, trimesh, fast-simplification.